

KARTIK SHARMA

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PROFILE

Full-stack developer with a strong foundation in Java, Data Structures & Algorithms, and RAG systems using LangChain. Focused on backend and system design fundamentals for Software Engineer roles.

EDUCATION

Lakshmi Narain College of Technology & Science

B.Tech in Computer Science Engineering, CGPA: 8.48/10

Sep 2023 – Present

Expected May 2027

Shivpuri Public School

Class 12 (Senior Secondary) – 90.6%

2023

Shivpuri Public School

Class 10 (Secondary) – 96.6%

2021

PROJECTS

AI-Powered RAG Q&A Chatbot (*In Development*) | *LangChain.js, Express.js, React, Milvus, Groq*

- Building a high-performance retrieval-augmented generation (RAG) pipeline with document ingestion, chunking, and binary-quantized vector embedding search, with an estimated 25–35ms retrieval latency.
- Designing a JavaScript-native RAG architecture (React + Express + LangChain.js + Milvus) using binary vector indexing and Hamming-distance search, projected to support efficient retrieval at scale.
- Integrating Groq's inference engine for the Kimi K2 model to serve low-latency, streamed generation responses via server-sent events.

Gamified Productivity Dashboard | *React.js, Node.js, Express.js, MongoDB*

- Built a full-stack productivity app with task management, daily planning, and gamification (points, leaderboards, rankings).
- Designed RESTful API endpoints for frontend-backend communication, handling endpoints across task and user modules.
- Structured scalable MongoDB schemas anticipating future feature expansion (badges, streaks, social features).

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript

Frontend: React.js, HTML5, CSS3, SCSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Database: MongoDB (Schema Design, Indexing), Milvus (Vector Search)

AI/GenAI: LangChain, LangGraph, RAG, Groq

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Tools: Git, GitHub, Postman

COURSES

Data Structures and Algorithms Using Java – *Infosys Springboard*

Software Engineering and Agile Software Development – *Infosys Springboard*